

Module 2 Word Cards

Five decks, 100 words, built from the words in the Module 2 crosswords (Bio Tool #21). One deck per table of three or four. Print single-sided, cut on the dashed lines, keep each deck in its own envelope.

Deck	Words	Learning targets	Cards
D1	Cell Theory and Two Kinds of Cells	LT 2.1, 2.2	22
D2	Organelles at Work, Levels and Homeostasis	LT 2.2, 2.3	24
D3	The Cell Membrane	LT 2.4	16
D4	Transport and Tonicity	LT 2.5	23
D5	Bridges Back: Cells and the Molecular Core	Spiral: Units 1, 3, 4 and 9	15

Four rounds, about five minutes each

1. **Match.** Lay the meaning cards face up. Stack the word cards face down. Take turns turning over a word and reading it aloud; the group agrees on its meaning before anyone picks the pair up.
2. **Sort.** Split the matched pairs into three to five piles and give each pile a name. Every pile name needs a reason someone at the table can say out loud.
3. **Say it without saying it.** One person draws a word card and describes it without using the word or any part of it. The first person to name it keeps the card.
4. **Connect.** Each person draws two word cards and says one true sentence that joins them. The group checks the sentence against the meanings.

★ **New words.** Ten cards carry a star. These words help on the SOL but have not come up in our Module 2 lessons yet; read their meanings together before you play. aquaporin (D3) · cellular respiration (D5) · contractile vacuole (D4) · glycoprotein (D3) · guard cells (D2) · interdependent (D2) · plasmolysis (D4) · receptor (D3) · turgor pressure (D4) · virus (D5).

D5 reaches back to Module 1 and ahead to Units 3 and 9

D5 brings back Module 1 words (polar, monomer, nucleotide, enzyme) and previews Unit 3 and Unit 9 words, each tied to a part of the cell. These are the words students most often lose by June.

D1 · Cell Theory and Two Kinds of Cells · the words

LT 2.1, 2.2 · Teal bar = word. ★ = new word. Cut on the dashed lines.

D1 hypothesis	D1 theory	D1 law	D1 tenet
D1 microscope	D1 technology	D1 evidence	D1 observation
D1 prediction	D1 cell division	D1 metabolism	D1 hereditary information
D1 prokaryote	D1 eukaryote	D1 bacteria	D1 organelle
D1 cytoplasm	D1 cell wall	D1 chloroplast	D1 vacuole
D1 centrioles	D1 flagella	D1 <i>blank: add a word from class</i>	D1 <i>blank: add a word from class</i>

D1 • Cell Theory and Two Kinds of Cells • what they mean

LT 2.1, 2.2 • Terracotta bar = meaning. Cut on the dashed lines.

D1 What you notice with your senses or an instrument, before you explain anything: "the seaweed slide shows boxes with green dots."	D1 How all new cells arise: one existing cell splits into two.	D1 Long whip-like tails that lash or spin to push a cell along. A sperm cell has one; many bacteria have several.	D1 A single cell whose DNA sits loose inside it, with no nucleus and no membrane-wrapped compartments. Bacteria are this kind.
D1 Green organelle that captures light energy to make sugar. In a spinach leaf, not in your cheek cells.	D1 One of the core statements of the cell theory, such as "all cells come from cells that already exist."	D1 An if-then statement of what you expect to see when you test an idea: if new cells come only from old ones, a sealed sterile broth stays clear.	D1 New tools and methods. Better lenses, then electron beams, let scientists see more inside cells and add to what they knew.
D1 A cell that keeps its DNA inside a nucleus and has many membrane-wrapped organelles. Plants, animals, fungi and seaweed.	D1 Storage sac for water and wastes. In a plant cell one large central one presses outward and keeps the plant from wilting.	D1 Stiff layer outside the membrane of plants, fungi and most bacteria. It is why lettuce is crisp; animal cells have none.	D1 Describes what will happen, every time, such as a dropped ball falling, without explaining why.
D1 The best-known single-celled organisms with no nucleus. A few cause strep throat; far more help digest your food.	D1 A broad explanation of why or how something happens, backed by many tests from many scientists. New evidence can still revise it.	D1 Instructions carried in DNA that pass from a cell to the two new cells when it splits.	D1 A pair of short tube bundles that help an animal cell divide; plant cells manage without them.
D1 Data and observations that support or refute a claim. A statement about cells is accepted only after many scientists find it again.	D1 "Little organ": a structure inside a cell with its own job, often wrapped in its own membrane.	D1 A proposed explanation for one question that can be tested. If no experiment could ever check it, it is not one.	D1 The instrument that first showed cork, pond water and leaves to be made of tiny boxes. Without it, nobody could have known cells exist.
D1 All the chemical reactions an organism needs to stay alive. A modern tenet says they happen inside cells.	D1 The jelly-like fluid, and everything floating in it, that fills a cell. Every kind of cell has it.	D1 <i>blank: write its meaning</i>	D1 <i>blank: write its meaning</i>

D2	D2	D2	D2
nucleus	ribosome	rough ER	Golgi apparatus
D2	D2	D2	D2
vesicle	lysosome	mitochondria	cytoskeleton
D2	D2	D2	★D2
protein	ATP	insulin	interdependent
D2	D2	D2	D2
unicellular	multicellular	specialized	tissue
D2	D2	D2	D2
organ system	organism	homeostasis	stomata
★D2	D2	D2	D2
guard cells	heart rate	sweat	roots

D2 · Organelles at Work, Levels and Homeostasis · what they mean

LT 2.2, 2.3 · Terracotta bar = meaning. Cut on the dashed lines.

D2	D2	D2	D2
Network of protein fibers and tubes that holds a cell's shape and serves as tracks for moving cargo.	Underground parts that take up water and minerals and help a plant keep its water balance steady.	Folded membrane highway studded with ribosomes, where new protein chains are threaded in and sent onward.	Hormone built by pancreas cells and shipped out to lower blood sugar: nucleus, ribosome, rough ER, Golgi, vesicle, out the door.
D2	D2	D2	D2
Stack of flattened sacs that modifies, sorts and labels proteins before they leave: the shipping center.	Keeping conditions inside steady, such as temperature, water and blood sugar, while conditions outside change.	Organelles with a folded inner membrane that release the energy stored in sugar as ATP. A muscle cell packs in thousands.	Made of one cell that must carry out every job of life by itself, like a pond protist with an eyespot and a whip-tail.
D2	D2	D2	D2
A group of similar cells doing the same job, such as muscle or bone; the level just above the cell.	The pair of curved cells around each leaf pore; they swell to open it and go limp to close it.	Beats per minute. It climbs during a sprint so blood delivers more oxygen, then falls back as you rest.	Depending on one another to work. A cell's organelles are this way: block one stop on the protein route and the rest back up.
D2	D2	D2	D2
Built for one job: a nerve cell grows long branches, and a red blood cell gives up its nucleus to carry more oxygen.	Small energy-carrying molecule made in mitochondria from sugar; a cell spends it to do work.	Chain of amino acids that the cell builds, folds and ships. Enzymes, antibodies and many hormones are examples.	Small membrane bubble that carries cargo from one compartment to the next, or out of the cell.
D2	D2	D2	D2
A whole living thing, the top level of organization. A single pond cell counts, and so does a whale.	Tiny pores on a leaf that let carbon dioxide in and water vapor out; closing them saves water on a hot day.	Several organs working as a team, like the heart, vessels and blood that deliver oxygen to every cell.	Tiny bead, in every living cell, that links amino acids into a protein chain.
D2	D2	D2	D2
Made of many cells that divide up the work. A seaweed frond and an oak tree both are.	Salty water your skin releases on a hot day; it cools you as it evaporates.	Control center wrapped in a double membrane. It holds the DNA instructions for everything the cell builds.	Sac of digestive enzymes that breaks down worn-out parts and swallowed bacteria: the recycling crew.

D3 · The Cell Membrane · the words

LT 2.4 · Teal bar = word. ★ = new word. Cut on the dashed lines.

D3 cell membrane	D3 phospholipid	D3 phospholipid bilayer	D3 hydrophilic
D3 hydrophobic	D3 cholesterol	D3 integral protein	D3 peripheral protein
D3 channel protein	D3 transport protein	★ D3 aquaporin	★ D3 receptor
★ D3 glycoprotein	D3 carbohydrate chains	D3 fluid mosaic model	D3 selectively permeable

D3 · The Cell Membrane · what they mean

LT 2.4 · Terracotta bar = meaning. Cut on the dashed lines.

D3	D3	D3	D3
"Water-fearing": describes the fatty tails hidden in the middle of the membrane, away from water.	A protein that passes all the way through the membrane, from one side to the other.	Lipid tucked between the tails in animal membranes; it keeps the membrane from turning too stiff or too runny.	Two sheets back to back: tails tucked in the middle, heads facing water on both sides.
D3	D3	D3	D3
Protein a signal molecule fits like a key in a lock, so the cell can receive a message such as a hormone.	A protein that sits on only one surface of the membrane and does not cross it.	The membrane's parts drift and shift like boats on a lake: a patchwork that is never frozen in place.	Molecule with a phosphate head and two fatty-acid tails; thousands line up to build the cell's boundary.
D3	D3	D3	D3
"Water-loving": describes the phosphate heads that face the watery inside and outside of the cell.	Short sugars sticking out from the outer surface of the membrane, used for recognition between cells.	Thin, flexible boundary around every cell that controls what enters and what leaves.	A channel protein built only for water. Kidney and root cells are packed with them.
D3	D3	D3	D3
Any membrane protein that moves a substance across: a tunnel it flows through, or a carrier that changes shape to pass it.	Lets some substances through and keeps others out.	A protein tunnel that lets one kind of ion, or water, slip through the membrane.	A membrane protein with a sugar chain attached; the chains work like name tags so cells recognize each other.

D4 · Transport and Tonicity · the words

LT 2.5 · Teal bar = word. ★ = new word. Cut on the dashed lines.

D4 diffusion	D4 osmosis	D4 facilitated diffusion	D4 passive transport
D4 concentration	D4 concentration gradient	D4 equilibrium	D4 solute
D4 solvent	D4 rate of diffusion	D4 hypertonic	D4 hypotonic
D4 isotonic	D4 burst	★ D4 plasmolysis	★ D4 turgor pressure
★ D4 contractile vacuole	D4 active transport	D4 protein pump	D4 endocytosis
D4 exocytosis	D4 phagocytosis	D4 pinocytosis	D4 <i>blank: add a word from class</i>

D4 • Transport and Tonicity • what they mean

LT 2.5 • Terracotta bar = meaning. Cut on the dashed lines.

D4	D4	D4	D4
Moving particles from low to high, uphill, which costs the cell ATP.	Downhill and free of cost, but through a protein doorway, the way glucose enters a red blood cell.	The substance that dissolves, such as salt or sugar stirred into water.	"Cell drinking": the membrane takes in tiny droplets of fluid and whatever is dissolved in them.
D4	D4	D4	D4
Surroundings that match the cell, so water moves in and out at the same rate and the cell keeps its size.	How much of a substance is packed into a given volume; 3 grams of salt per liter is higher than 1.	Surroundings with more dissolved particles than the cell; the cell loses water and shrinks, like celery going limp in salt water.	The membrane folds in around something outside the cell and pinches off a bubble, bringing it inside.
D4	D4	D4	D4
Water pressing the membrane out against the cell wall; it keeps a watered plant standing stiff.	"Cell eating": a white blood cell wraps around a bacterium and swallows it whole.	How fast particles spread. A steeper difference from high to low, or a warmer temperature, makes it faster.	A vesicle merges with the membrane and spills its contents outside; how a pancreas cell releases insulin.
D4	D4	D4	D4
Surroundings with fewer dissolved particles than the cell; water rushes in, as when a saltwater seaweed lands in fresh water.	A difference in amount from one place to another. Particles slide down it, from high to low.	Any movement across the membrane that costs the cell no energy, because particles go from crowded to less crowded.	Pump-like sac in a freshwater protist that squeezes out the extra water that keeps flowing in.
D4	D4	D4	D4
What an animal cell, with no wall to hold it, can do when too much water rushes in.	Particles spreading from where they are crowded to where they are scarce, like food coloring through still water.	Membrane protein that spends ATP to push ions uphill, against their gradient; the sodium-potassium one runs in every nerve cell.	The substance that does the dissolving; in living things it is almost always water.
D4	D4	D4	D4
A plant cell's membrane pulling away from its wall as the cell loses water in salty surroundings.	Water crossing a membrane toward the side with more dissolved particles; why a slug shrivels under table salt.	The point where particles are spread evenly, so movement one way is balanced by movement the other way.	<i>blank: write its meaning</i>

D5polar	D5hydrogen bond	D5monomer	D5polymer
D5amino acid	D5nucleotide	D5nucleic acid	D5lipid
D5glucose	D5enzyme	D5chlorophyll	D5photosynthesis
★D5cellular respiration	D5DNA	★D5virus	D5blank: add a word from class

D5 · Bridges Back: Cells and the Molecular Core · what they mean

Spiral: Units 1, 3, 4 and 9 · Terracotta bar = meaning. Cut on the dashed lines.

D5	D5	D5	D5
The weak pull between a slightly positive hydrogen and a slightly negative atom nearby; it holds water molecules together.	One small building block. Many of them joined make a polymer.	Protein that speeds up one reaction without being used up; lysosomes are full of them.	Mitochondria use glucose and oxygen to make ATP, releasing carbon dioxide and water. Plants do it too.
D5	D5	D5	D5
Building block of DNA and RNA: a sugar, a phosphate and a nitrogen base.	A large molecule made of many repeating building blocks joined in a chain.	Not a cell: genetic material in a protein coat, with no ribosomes or metabolism of its own; it must use a host cell to copy itself.	Green pigment inside chloroplasts that absorbs light energy.
D5	D5	D5	D5
Having a slightly negative end and a slightly positive end, like water; why water sticks to itself and dissolves salt.	The cell's instructions, kept in the nucleus of a eukaryote and loose in the cytoplasm of a prokaryote.	Fats, oils and the phospholipids of every membrane; mixes poorly with water and stores energy long-term.	Building block of proteins; ribosomes link them in the order the DNA instructions give.
D5	D5	D5	D5
Polymer that stores and carries information: DNA in the nucleus, RNA on its way to the ribosome.	Simple sugar made in chloroplasts and broken down in mitochondria to make ATP.	Chloroplasts use light energy, carbon dioxide and water to make glucose and release oxygen.	<i>blank: write its meaning</i>